XIAOFAN CUI

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RESEARCH PROFILE

My research focuses on uncertainty-aware personal agents and adaptive human-AI interaction. I investigate how agents interpret ambiguous requests, learn context-dependent user preferences, and decide when to act or seek clarification. My work combines computational prototyping with empirical user research, using smart environments as an initial testbed.

EDUCATION

Shanghai Jiao Tong University, Shanghai, China **Sep 2024 - Mar 2027 (expected)**
Master of Mechanical Engineering
GPA: 3.7/4.0

Northeastern University, Shenyang, China **Sep 2019 - May 2024**
Bachelor of Architecture
Average Score: 81/100

PUBLICATIONS AND MANUSCRIPTS

Zhou, T., Xu, Y., **Cui, X.**, Qin, T., Guan, H., Deng, J., & Chen, Z. (2026). Knowledge and data co-driven Bayesian network for personalized user experience evaluation of smart product service system. *Advanced Engineering Informatics, 74*, 104674. https://doi.org/10.1016/j.aei.2026.104674

Cui, X., Xia, Y., Xu, Y., Yang, Y., Han, T., Cao, Y., & Zhou, T. (2026). When user requests are ambiguous: The role of response explainability in shaping user experience with LLM-based smart home assistants. *International Journal of Human-Computer Interaction*. **（Revised manuscript submitted to *International Journal of Human–Computer Interaction* following the second round of peer review）**

RESEARCH EXPERIENCE

Master's Thesis — Adaptive and Explainable LLM Agents for Ambiguous Interaction in Smart Environments **Mar 2026 - Present**
Master's Researcher and Project Lead
Investigated how LLM-based agents should respond to ambiguous human intent while balancing autonomous action, explanation, clarification, and user control in smart environments.

Study 1 — Effects of Request Ambiguity and Response Explainability

- **Research question:** Investigated how user request ambiguity and LLM response explainability jointly affect trust, satisfaction, and pleasure in smart-home interaction.
- **Method:** Conducted a 2 × 2 between-subjects scenario-based study with 211 valid participants, examining how request ambiguity and response explainability affected trust, satisfaction, and enjoyment.
- **Finding:** The results indicated that, under conditions of low response explainability, ambiguous user requests significantly reduced trust, satisfaction, and pleasure. However, enhanced response explainability significantly mitigated the adverse effects of request ambiguity on these three user-experience outcomes.

Study 2 — User Modelling and Uncertainty-Aware Interaction in Smart Environments

- **Research question:** Investigated how an LLM-based agent should adapt its level of autonomy when user requests are ambiguous, including when and how to act, explain, clarify, or offer alternatives.
- **Method:** Developed a closed-loop multi-agent system that conducts interactions in a controllable Unreal Engine smart-environment simulation, records user responses and preferences in persistent memory, updates an individual user model, and uses the learned model to adapt subsequent explanation decisions.
- **Finding:** Technical validation showed that the system could update individual preference profiles from recorded explanation requests, acceptance, rejection, and corrections, and use these profiles to adapt explanation strategies across turns and sessions. A formal user study evaluating the effects on trust, perceived control, and user experience is in preparation.

Researcher — AI-Assisted Dynamic User Demand Analysis for Product Design **Nov 2024 - May 2025**

Project Lead and Lead Researcher

- **Research question:** Investigated how time-stamped online reviews could be transformed from fragmented, static feedback into dynamic and actionable evidence for product-design prioritization.
- **Method:** Following data acquisition by collaborators, analyzed 2,000 automotive reviews comprising 3,674 sentence-level records through feature clustering, sentiment estimation, and time-series forecasting, and developed a three-agent workflow and visual analytics platform to translate demand trends into design priorities and improvement strategies.
- **Finding:** In a BYD Song PLUS case study, the approach identified legroom as the highest-priority spatial feature and surfaced recurring problems concerning body-size accommodation, full-occupancy space allocation, and ergonomic support. A case-based comparison indicated that the structured workflow produced more focused, prioritized, and traceable design recommendations than direct analysis with a standalone LLM.

Researcher — Women on the Wheel: Urban Cycling Service for Female University Students **Sep 2024 - Nov 2024**

User Research and Service Design Contributor

- Investigated how equipment access, route uncertainty, safety concerns, and gendered social pressures shape female university students' entry into and continued participation in urban cycling.
- Contributed to a mixed-method study combining desk research, semi-structured interviews, a questionnaire, user segmentation, personas, journey mapping, and stakeholder analysis; synthesized the findings into user scenarios and service-design opportunities.
- Found that cycling experience alone did not adequately explain users' needs: their initial motivations and interests also shaped information-seeking and participation, while equipment access, fragmented route and hazard information, and limited female-oriented support emerged as recurring barriers.
- Translated these insights into a digital–physical service concept integrating route discovery, hazard information, bicycle access, ride documentation, and community support for novice and experienced riders.

TEACHING EXPERIENCE

Teaching Assistant — Design Fundamentals II: Programming Fundamentals **Mar 2025 - May 2025**

School of Design, Shanghai Jiao Tong University

- Supported first-year undergraduate students throughout a 13-week Python programming course through classroom demonstrations, technical consultation, and assignment feedback.
- Graded assignments and examinations, guided design-oriented programming projects, and provided individual technical support.

SCHOLARSHIPS

Second-Class Academic Scholarship, Shanghai Jiao Tong University, 2025

Second-Class Academic Scholarship, Shanghai Jiao Tong University, 2026

RESEARCH AND TECHNICAL SKILLS

Machine Learning and NLP: Topic modelling; sentiment analysis; time-series modelling.

Agent Systems: Structured intent representation; contextual preference modelling; clarification and feedback mechanisms.

Research Methods: Scenario-based experiments; questionnaires; interviews; quantitative analysis.

Programming and Prototyping: Python; Unreal Engine 5; Unity; ESP32.

Spatial Design: Blender; Rhino; SketchUp; D5 Render; CAD

Languages: Chinese (native), English (IELTS 6.5)